Linux `tar` Command Notes

A **TAR file** is a collection of several files and directories in one file. This is a good way to create backups and archives.

🔹 Syntax

```
tar [options] [archive-file] [file or directories to be archived]
```

🔹 Options

Option	Description
`-c`	Create a new archive
`-f`	Specify archive file name
`-t`	Show the list of files in the tar file
`-v`	Show the progress (verbose mode) while creating or extracting
`-r`	Append files to the end of an archive
`-x`	Extract files from an archive
`-z`	Compress the tar file with **gzip** → creates `.tar.gz`
`-j`	Compress the tar file with **bzip2** → creates `.tar.bz2`

🔹 Examples

1. Create a new archive

```
tar -cvf archive.tar file1.txt file2.txt dir/
```

👉Creates `archive.tar` containing `file1.txt`, `file2.txt`, and `dir/`.

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2. List files inside an archive

```
tar -tvf archive.tar
```

👉Displays the list of files inside `archive.tar` without extracting.

Example Output:

```
-rw-r--r-- user/group  1024 2025-09-22 10:00 file1.txt
-rw-r--r-- user/group  2048 2025-09-22 10:05 file2.txt
drwxr-xr-x user/group     0 2025-09-22 10:10 dir/
```

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3. Extract files from an archive

```
tar -xvf archive.tar
```

👉Extracts all files from `archive.tar`.

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4. Append files to an existing archive

```
tar -rvf archive.tar newfile.txt
```

👉Adds `newfile.txt` to the existing `archive.tar`.

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5. Create gzip compressed tarball

```
tar -czvf archive.tar.gz file1.txt dir/
```

👉Creates `archive.tar.gz` using **gzip**.

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6. Create bzip2 compressed tarball

```
tar -cjvf archive.tar.bz2 file1.txt dir/
```

👉Creates `archive.tar.bz2` using **bzip2**.

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7. Extract compressed archives

- **Extract gzip (.tar.gz)**

```
tar -xzvf archive.tar.gz
```

- **Extract bzip2 (.tar.bz2)**

```
tar -xjvf archive.tar.bz2
```

🔹 Summary

- `tar` bundles multiple files/folders into a single archive.
- Use `-c` to create, `-x` to extract, `-t` to list.
- Use `-f` to specify filename.
- Use `-v` for verbose output.
- Use `-r` to append.
- Use `-z` for gzip compression (`.tar.gz`).
- Use `-j` for bzip2 compression (`.tar.bz2`).

👈🏻 **Remember:** - `tar` alone = just archives (no compression). - `tar + -z / -j` = archive + compress.